Abhinav Vimal

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Education

Degree/Certificate	Institution	Percentage/CGPA
B-Tech	IIT BHU, Varanasi	7.78
CBSE-XII	JNV, Unnao	87.6
CBSE-X	JNV, Unnao	91.20

Certifications/ Trainings

- NVIDIA Certified Associate – Generative AI & LLMs
- GSI NVIDIA Technologies Curriculum (2024, 2025)
- AI for All: From Basics to GenAI Practice
- NVIDIA AI Technical Curriculum (2024, 2025)

Overall Experience

Languages - Python, C++.

Statistical Tools – Pandas, Numpy, Mapper (TDA), SciPy.
Data Visualization Tools – Matplotlib, Seaborn, Plotly.
Imaging Tools – OpenCV , Pillow (PIL), Scikit-image, torchvision.
Video/Multimedia Processing Tools – FFmpeg, GStreamer.
Hardware Platforms - Nvidia Jetson (AGX Orin, Xavier, Nano, TX2), Nvidia dGPU machine (H200, H100, V100, A100 etc.), Intel Gaudi 3, Intel ASUS NUC 15 Pro+ Edge device, AMD EPYC CPUs.
Nvidia Tech Stack – Nemo, NIM, DeepStream, TensorRT, TAO, Riva, Kaolin etc.
Intel Tech Stack – OpenVINO, DL Streamer, Optimum-Habana.

ML/DL Tools and Libraries – Tensorflow, Pytorch, Scikit-learn.
MLOps Tools and Technologies – Mlflow, Airflow, DVC & CML, MLEM, Git, Github, Github actions, Feast, Docker etc.
AWS Tools and Technologies – EC2, ECS, S3, SageMaker, Bedrock, ECR, Redshift, DynmoDB, Lambda.
Generative AI Tools and Technologies – Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), PEFT (LoRA, QLoRA), LLM Inference and Serving, vLLM, Llama.cpp, Ollama, TensorRT-LLM, LLM Fine-tuning & Quantization, DeepSpeed, Diffusion Models, Prompt Engineering, LangChain, LlamaIndex, LLM Evaluation, Guardrails, LLM Agents, LangGraph, Agentic AI.

- 4.5+ years of hands-on experience in Artificial Intelligence, covering Machine Learning, Deep Learning, Computer Vision, MLOps, LLMOps, and Generative AI systems.
- Experience in computer vision applications, including image classification, object detection, multi-object tracking, segmentation, and real-time video analytics at the edge.
- Hands-on experience building MLOps CI/CD pipelines on AWS SageMaker for model training, validation, and workflow automation.
- Experience with ML pipeline orchestration and automation, including feature engineering, experiment tracking, model versioning, and deployment workflows.
- Experience with Trustworthy AI practices, including fairness, explainable AI (XAI), secure AI, and ethical AI.
- Hands-on experience building Generative AI applications, including RAG, multimodal RAG and LLM-powered systems.
- Hands-on experience designing and building agentic AI systems using LangGraph, including graph-based orchestration, stateful workflows, tool-calling, human-in-the-loop and multi-step planning/execution.
- Experience in LLM fine-tuning and inference optimization using PEFT techniques (LoRA, QLoRA), DeepSpeed, TensorRT-LLM, and hardware-aware optimizations.
- Developed GenAI SDKs and applications using LangChain, LlamaIndex, and Semantic Kernel frameworks.
- Experience with containerization and Docker-based workflows for ML and AI services.

PROJECTS

Company Name	Roles & Responsibilities
LTM	**Project: Transit Watch – Edge AI Metro Safety & Video Analytics**

<table>
<tr><td></td><td>Key Responsibilities:

• Developed a real-time edge AI video analytics solution for metro safety and operational monitoring using live CCTV streams.
• Built computer vision pipelines to detect events such as yellow line crossing, intrusion, unattended objects, wrong-direction movement etc.
• Designed and optimized edge deployments tailored to specific hardware architectures, including Intel Core Ultra–based ASUS NUC Pro 15+ and NVIDIA Jetson AGX Orin device.
• Leveraged Intel OpenVINO to utilize CPU, GPU, and NPU acceleration on Intel platform, and NVIDIA TensorRT for GPU-accelerated, low-latency inference on Jetson.

Tools – Intel OpenVINO, Intel DL Streamer, NVIDIA TensorRT, DeepStream SDK, FFmpeg, GStreamer, PostgreSQL.
Technology – Edge AI, Real-time Computer Vision, Video Analytics, Hardware-Aware Optimization, Low-latency Inference, Edge Deployment.</td></tr>
<tr><td>LTM</td><td>Project: PAN 2.0 – AI-Based Identity Deduplication & Verification

Key Responsibilities:

• Built an API-driven deduplication pipeline to validate PAN applications using photo, signature, and textual attributes.
• Implemented face similarity validation using DeepFace and signature similarity using deep learning–based embeddings with cosine similarity matching.
• Designed multimodal verification workflows combining image and text signals, generating similarity scores for downstream decisioning and reporting.
• Deployed containerized inference and validation services on AWS ECS.

Tools – DeepFace, TensorFlow, PyTorch, FastAPI, Docker, AWS EC2, ECR, ECS, Git, GitHub.
Technology – Face Recognition, Image Embedding & Matching, Vector Similarity Search, Cosine Similarity Scoring, Multi-modal Feature Extraction, Threshold-based Decisioning, Cloud-native AI Deployment.</td></tr>
<tr><td>LTM</td><td>Project: LLM Factory – Hardware-Aware LLM Fine-Tuning, Optimization & Serving Platform

Key Responsibilities:

• Built an end-to-end platform for LLM fine-tuning, optimization, and serving across diverse hardware environments.
• Integrated multiple LLM fine-tuning frameworks including NVIDIA NeMo, Unsloth, Intel Optimum-Habana, and InstructLab.
• Designed training pipelines with evaluation, logging, checkpointing, and resource monitoring.
• Enabled flexible LLM serving using NVIDIA NIM, TensorRT-LLM, OpenVINO, vLLM, llama.cpp, and Ollama

Frameworks/Tools – NVIDIA NeMo, Hugging Face, Unsloth, Intel Optimum-Habana, InstructLab.
Technology – LLM Fine-Tuning, Hardware-Aware Optimization, Quantization, PEFT, Model Evaluation, LLM Serving.</td></tr>
<tr><td>HCLTech</td><td>Intel GenOps Project: Automated Data Crawler, Ingestion, and Retrieval Framework

Key Responsibilities:

• Designed and implemented source-specific crawlers to efficiently extract data from multiple data sources while complying with access and security protocols.
• Developed a Data Ingestion pipeline to facilitate the transfer, organization, and storage of collected data, ensuring seamless integration with retrieval processes.
• Built an efficient retrieval pipeline capable of delivering accurate, context-aware responses to user queries with low latency and high relevance.

Tools and Technology used: Apache Airflow, MinIO, PostgreSQL, Langchain and Llamaindex SDK, RAG.</td></tr>
<tr><td>HCLTech</td><td>Project: Dynamic MLOps best practices incorporating Responsible AI functionalities</td></tr>
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	Key Responsibilities: • Implementing and optimizing MLOps (managing the lifecycle of ML/DL models, including development, deployment, and maintenance). • Ensuring deployed model's fairness, explainability, security and robustness. • Implementing and optimizing LLMOps (managing the lifecycle of LLM and LLM-powered applications, including development, deployment, and maintenance). **Tools** – Mlflow , DVC, CML, Git, Github, Github actions, AWS S3 and EC2. **Technology** – Data Ingestion, Data Processing, Model Fine-tuning and Monitoring, PEFT (LoRA, QLoRA), Prompt Engineering, Distributed Training.